

Revised Generation Interconnection System Impact Study Report

For Queue Project Q17-0771

Jack McDonough 138 kV
June 02, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0771. The Interconnection Customer proposes a 466.0 MW battery energy storage facility interconnecting at the Jack McDonough 138 kV substation in the Maryland area. The request was received on April 06, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 225 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 41 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$147,137,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Low Gap-Jack McDonough 138 KV line	Artesia Country Club-Air Force Base 345 KV	178.4	355	48.8
Canovanas Pueblo 2401-Low Gap 138 KV Line	Artesia Country Club-Air Force Base 345 KV	174.8	244	31.8
Low Gap Substation-Flint River 138 KV line	Low Gap-Canovanas Pueblo 2401 138 KV	158.0	182	29.4
Trumann-Rossmoor S 138 KV	Air Force Base-Tiger Bay 345 KV	157.7	229	5.4
Brighton-Drum 138 KV	Artesia Country Club-Air Force Base 345 KV	156.2	312	6.1
Jack McDonough-Lamont 138 KV	Artesia Country Club-Air Force Base 345 KV	154.7	433	51.2
Lamont-Dundee 138 KV Line	Shepherdstown-Peachland Substation 138 KV	150.6	215	16.9
Metro Ctr-Dundee 138 KV L	Cameron-Athalmer Tap 138 KV	146.5	333	9.5
Rockville-Phillips Pump 1 138 KV	Tramway-Rossmoor S 138 KV	145.4	162	5.8
Lamont-University City 138 KV	Clark County Substation-Kearneyville 345 KV	145.0	291	11.0
Hollister-El Sobrante Substation 138 KV	E Hagerstown-Pineville 138 KV	140.3	411	8.6
Montgomery Village-Bondurant 138 KV L	Montgomery Village-Bondurant 138 KV Ckt 2	138.6	361	7.5
University City-Richford 138 KV	Clark Couty-Kearneyville 345 KV	138.4	330	11.0

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
M105 Substation-Wyoming 138 KV	W Waynesboro-Chambersburg 345 KV	137.0	394	5.0
Air Force Base Substation-Rancho Substation 138 KV	Air Force Base-Artesia Country Club 345 KV	136.3	202	6.1
Sun City-Richford 138 KV	Clark County-Kearneyville 345 KV	136.1	345	11.0
Pineville-Innerarity 138 KV Line	Pineille-E Hagerstown 138 KV	135.0	458	8.6
Maryland Solar-Bellsfork 138 KV	East Hagerstown-Pineville 138 KV	135.0	449	8.6
Sun City-Peachland Substation 138 KV	Kearneyville-Clark County 345 KV	134.8	355	11.0
Rockville-Scarcello 138 KV	Tramway-Rossmoor South 138 KV line	134.3	194	5.8
Shepherdstown-Peachland 138 KV	Kearneyville-Clark County 345 KV	134.2	362	11.0
Bondurant-Tramway Substation 138 KV Line 1	Bondurant-Tramway 138 KV	134.2	360	7.5
Cheatham-Canovanas Pueblo 201 138 KV	Air Force Base-Artesia Country Club 345 KV	133.6	290	31.8
Antietam-Pineville 138 KV	Pineville-West Wayneboro 345 KV	133.6	398	6.7
Tramway-Scarcello 138 KV	Tramway-Rossmoor South 138 KV	133.3	245	5.8
Tramway-Rossmor S 138 KV	Tiger Bay-Air Force Base 345 KV Line	132.5	313	5.4
Messick Substation-Pluess 138 KV L	Metr Center-Dundee 138 KV	132.5	438	7.6
Cameron-Pluess 138 KV Line	Dundee-Metro Ctr 138 KV	131.7	455	7.6
Athalmer Tap-Skykomish 138 KV	Innerarity-Pineville 138 KV	131.2	220	6.4
Tramway-Bondurant 138 KV #2	Bondurant-Tramway 138 KV Circuit 1	131.0	368	7.5
Cameron-Athalmer Tap 138 KV	Dundee-Metro Ctr 138 KV	130.8	468	7.6
Athalmer Tap-East Hagerstown Substation 138 KV L	Innerarity-Pineville 138 KV Line	129.3	568	9.9
Messick Substation-Five hannels 138 KV L	Metro Ctr-Dundee 138 KV	128.7	430	7.6
Morgan Falls-Four Hills 138 KV	Hutchinson Plant 1-Watford City 138 KV	128.2	237	6.9
Bellsfork-El Sobrante 138 KV	Pineville-East Hagerstown 138 KV L	127.9	468	8.6
Innerarity-Maryland Solar 138 KV L	Pineville-East Hagerstown 138 KV	127.2	476	8.6
Pineville-Est Hagerstown 138 KV Line	Innerarity-Pineville 138 KV Line	126.9	605	9.9
Montgoery Village-Bondurant 138 KV Ckt 2	Montgomery Village-Bondurant Substation 138 KV	126.7	395	7.5
Reid-Clawson 138 KV	W Waynesboro-Pinville 345 KV	126.7	476	6.7
Middlebrook-Rancho 138 KV L	Artesia Country Club-Air Force Base 345 KV line	125.7	238	6.1
Artesia Country Club-Clark County Substation 345 KV	Jack McDonough-Low Gap 138 KV	109.7	1,291	14.2

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Jack McDonough-Low Gap 138 KV	Rebuild line with higher-capacity conductor	\$1,717,000	60% (\$1,033,000)
Canovanas Pueblo 2401-Low Gap 138 KV Line	Rebuild line with higher-capacity conductor	\$9,492,000	88% (\$8,362,000)
Low Gap-Flint Riv 138 KV L	Rebuild line with higher-capacity conductor	\$7,667,000	100% (\$7,667,000)
Rossmoor South-Trumann 138 KV	Rebuild line with higher-capacity conductor	\$7,145,000	100% (\$7,145,000)
Brighton-Drum 138 KV	Rebuild line with higher-capacity conductor	\$6,164,000	100% (\$6,164,000)
Lamont-Jack McDonough 138 KV Line	Rebuild line with higher-capacity conductor	\$4,311,000	61% (\$2,644,000)
Lamont-Dundee 138 KV Line	Rebuild line with higher-capacity conductor	\$4,164,000	90% (\$3,748,000)
Dundee-Metro Ctr 138 KV	Rebuild line with higher-capacity conductor	\$12,912,000	29% (\$3,754,000)
Rockville Substation-Phillips Pump 1 138 KV	Rebuild line with higher-capacity conductor	\$2,248,000	100% (\$2,248,000)
Lamont-University City 138 KV	Rebuild line with higher-capacity conductor	\$16,118,000	90% (\$14,585,000)
Hollister-El Sobrante Substation 138 KV	Rebuild line with higher-capacity conductor	\$4,082,000	25% (\$1,011,000)
Montgomery Village-Bondurant 138 KV L	Rebuild line with higher-capacity conductor	\$6,934,000	100% (\$6,934,000)
University City-Richford 138 KV	Rebuild line with higher-capacity conductor	\$4,886,000	90% (\$4,421,000)
Wyoming-M105 138 KV line	Rebuild line with higher-capacity conductor	\$1,080,000	15% (\$165,000)
Air Force Base Substation-Rancho Substation 138 KV	Rebuild line with higher-capacity conductor	\$767,000	100% (\$767,000)
Richford-Sun City Substation 138 KV Line	Rebuild line with higher-capacity conductor	\$12,254,000	90% (\$11,088,000)
Innerarity-Pineville 138 KV	Rebuild line with higher-capacity conductor	\$2,398,000	25% (\$594,000)
Maryland Solar-Bellsfork 138 KV	Rebuild line with higher-capacity conductor	\$4,447,000	25% (\$1,101,000)
Peachland-Sun City 138 KV Line	Rebuild line with higher-capacity conductor	\$3,808,000	90% (\$3,446,000)
Rockville-Scarcello 138 KV	Rebuild line with higher-capacity conductor	\$959,000	100% (\$959,000)
Shepherdstown-Peachland 138 KV line	Rebuild line with higher-capacity conductor	\$2,600,000	90% (\$2,353,000)
Bondurant-Tramway Substation 138 KV Line 1	Rebuild line with higher-capacity conductor	\$1,267,000	100% (\$1,267,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Cheatham-Canovanas Pueblo 201 138 KV	Rebuild line with higher-capacity conductor	\$4,703,000	88% (\$4,143,000)
Antietam-Pineville 138 KV	Rebuild line with higher-capacity conductor	\$4,233,000	18% (\$740,000)
Tramway-Scarcello 138 KV	Rebuild line with higher-capacity conductor	\$1,517,000	100% (\$1,517,000)
Tramway-Rossmor S 138 KV	Rebuild line with higher-capacity conductor	\$1,443,000	100% (\$1,443,000)
Messick-Pluess 138 KV	Rebuild line with higher-capacity conductor	\$6,962,000	20% (\$1,372,000)
Cameron-Pluess 138 KV Line	Rebuild line with higher-capacity conductor	\$2,391,000	20% (\$471,000)
Athalmer Tap-Skykomish 138 KV L	Rebuild line with higher-capacity conductor	\$7,132,000	100% (\$7,132,000)
Bondurant-Tramway 138 KV	Rebuild line with higher-capacity conductor	\$2,195,000	100% (\$2,195,000)
Cameron-Athalmer Tap 138 KV	Rebuild line with higher-capacity conductor	\$3,916,000	20% (\$772,000)
Athalmer Tap-East Hagerstown Substation 138 KV	Rebuild line with higher-capacity conductor	\$3,378,000	23% (\$775,000)
Messick-Five Channels Substation 138 KV line	Rebuild line with higher-capacity conductor	\$3,337,000	20% (\$658,000)
Morgan Falls-Four Hills 138 KV	Rebuild line with higher-capacity conductor	\$18,977,000	100% (\$18,977,000)
Bellsfork Substation-El Sobrante 138 KV	Rebuild line with higher-capacity conductor	\$3,579,000	25% (\$886,000)
Maryland Solar-Innerarity 138 KV	Rebuild line with higher-capacity conductor	\$969,000	25% (\$240,000)
Pineville-Est Hagerstown 138 KV Line	Rebuild line with higher-capacity conductor	\$4,553,000	23% (\$1,044,000)
Montgomery Village-Bondurant 138 KV Ckt 2	Rebuild line with higher-capacity conductor	\$9,009,000	100% (\$9,009,000)
Reid-Clawson 138 KV	Rebuild line with higher-capacity conductor	\$7,547,000	18% (\$1,320,000)
Middlebrook-Rancho 138 KV	Rebuild line with higher-capacity conductor	\$1,455,000	100% (\$1,455,000)
Artesia Country Club-Clark County 345 KV	Reconductor line	\$19,403,000	8% (\$1,532,000)
	Total allocated to project		\$147,137,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.